

Venus Labs - Executor Contract

Security Assessment

CertiK Assessed on Apr 27th, 2026





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Venus Labs - Executor Contract

The security assessment was prepared by Certik.

Executive Summary

TYPES

Lending

ECOSYSTEM

Binance Smart Chain
(BSC)

METHODS

Manual Review, Static Analysis

LANGUAGE

Solidity

TIMELINE

Preliminary comments published on 04/23/2026

Final report published on 04/27/2026

Vulnerability Summary



2

Total Findings

2

Resolved

0

Partially Resolved

0

Acknowledged

0

Declined



0 Centralization

Centralization findings highlight privileged roles & functions and their capabilities, or instances where the project takes custody of users' assets.



0 Critical

Critical risks are those that impact the safe functioning of a platform and must be addressed before launch. Users should not invest in any project with outstanding critical risks.



0 Major

Major risks may include logical errors that, under specific circumstances, could result in fund losses or loss of project control.



1 Medium

1 Resolved



Medium risks may not pose a direct risk to users' funds, but they can affect the overall functioning of a platform.



1 Minor

1 Resolved



Minor risks can be any of the above, but on a smaller scale. They generally do not compromise the overall integrity of the project, but they may be less efficient than other solutions.



0 Informational

Informational errors are often recommendations to improve the style of the code or certain operations to fall within industry best practices. They usually do not affect the overall functioning of the code.

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Disclaimer

CODEBASE | VENUS LABS - EXECUTOR CONTRACT

Repository

The changes introduced in <https://github.com/VenusProtocol/venus-periphery/pull/61>

Commit

<https://github.com/VenusProtocol/venus-periphery/tree/5a2d915e464c6d6ef53912cccaa6818299808a89/>

AUDIT SCOPE | VENUS LABS - EXECUTOR CONTRACT

CertiKProject/certik-audit-projects

 Executor/Executor.sol

 Executor/IExecutor.sol

 Interfaces/IVToken.sol

APPROACH & METHODS | VENUS LABS - EXECUTOR CONTRACT

This audit was conducted for Venus Labs to evaluate the security and correctness of the smart contracts associated with the Venus Labs - Executor Contract project. The assessment included a comprehensive review of the in-scope smart contracts. The audit was performed using a combination of Manual Review and Static Analysis.

The review process emphasized the following areas:

- Architecture review and threat modeling to understand systemic risks and identify design-level flaws.
- Identification of vulnerabilities through both common and edge-case attack vectors.
- Manual verification of contract logic to ensure alignment with intended design and business requirements.
- Dynamic testing to validate runtime behavior and assess execution risks.
- Assessment of code quality and maintainability, including adherence to current best practices and industry standards.

The audit resulted in findings categorized across multiple severity levels, from informational to critical. To enhance the project's security and long-term robustness, we recommend addressing the identified issues and considering the following general improvements:

- Improve code readability and maintainability by adopting a clean architectural pattern and modular design.
- Strengthen testing coverage, including unit and integration tests for key functionalities and edge cases.
- Maintain meaningful inline comments and documentations.
- Implement clear and transparent documentation for privileged roles and sensitive protocol operations.
- Regularly review and simulate contract behavior against newly emerging attack vectors.

FINDINGS | VENUS LABS - EXECUTOR CONTRACT



2

Total Findings

0

Critical

0

Centralization

0

Major

1

Medium

1

Minor

0

Informational

This report has been prepared for Venus Labs to identify potential vulnerabilities and security issues within the reviewed codebase. During the course of the audit, a total of 2 issues were identified. Leveraging a combination of Manual Review & Static Analysis the following findings were uncovered:

ID	Title	Category	Severity	Status
VLE-01	<code>handleBorrowCapExceeding()</code> And <code>handleSupplyCapExceeding()</code> Rely On Accrual-Triggering Reads Before Executing Emergency Actions, Making The Executor Path Brittle On Accrual Failure	Volatile Code	Medium	● Resolved
VLE-02	<code>handleSupplyCapExceeding()</code> Can Roll Back A Successful Supply Pause If The Subsequent Collateral-Factor Reduction Reverts	Volatile Code	Minor	● Resolved

VLE-01 `handleBorrowCapExceeding()` And `handleSupplyCapExceeding()` Rely On Accrual-Triggering Reads Before Executing Emergency Actions, Making The Executor Path Brittle On Accrual Failure

Category	Severity	Location	Status
Volatile Code	● Medium	Executor/Executor.sol: 131~132, 152~153	● Resolved

Description

Both emergency-condition handlers perform accrual-triggering reads before executing the safety action through `EBRAKE`.

In `handleSupplyCapExceeding(address)`, the contract calls `IVToken(market).exchangeRateCurrent()` while computing underlying supply before reaching `EBRAKE.pauseSupply(market)` and `EBRAKE.decreaseCF(market, 0)`. In `handleBorrowCapExceeding(address)`, it calls `IVToken(market).totalBorrowsCurrent()` before reaching `EBRAKE.pauseBorrow(market)`. These calls accrue interest and can revert if the market cannot complete accrual successfully, such as under excessive borrow-rate or other accrual-failure conditions.

If accrual reverts, the executor helper reverts before executing the corresponding emergency action, so this path becomes unavailable exactly when the market is already in a stressed state. The protocol still retains alternative emergency controls via the Comptroller's ACM-gated administrative pause functions, so the issue is a brittleness in the executor helper path rather than a protocol-wide loss of emergency response capability.

Scenario

1. An authorized caller invokes `handleSupplyCapExceeding(market)` or `handleBorrowCapExceeding(market)` for a market whose VToken cannot complete interest accrual successfully.
2. `exchangeRateCurrent()` or `totalBorrowsCurrent()` reverts before the function reaches the `EBRAKE` emergency action.
3. The executor helper cannot pause supply, zero CF, or pause borrowing through this path.
4. ACM-authorized operators can still use the Comptroller's direct administrative controls to pause actions.

Recommendation

Avoid making the safety-critical action depend entirely on an accrual-triggering read. If a non-accruing view is acceptable for the breach check, prefer that path. Otherwise, wrap the accrual-triggering call in `try/catch` and handle failure explicitly so the system can either fall back to an alternative emergency path or emit a clear failure signal instead of relying on a brittle revert. If the current behavior is intentional, document that these executor handlers are best-effort validated helpers and that operators must use direct Comptroller pause controls when accrual is broken.

I Alleviation

[Venus Labs, 04/27/2026]: The team heeded the advice and resolved the issue in <https://github.com/VenusProtocol/venus-periphery/pull/64/changes/b0454a9603c250c85e0411a507b63c8a0fc72b60>.

VLE-02 `handleSupplyCapExceeding()` Can Roll Back A Successful Supply Pause If The Subsequent Collateral-Factor Reduction Reverts

Category	Severity	Location	Status
Volatile Code	Minor	Executor/Executor.sol: 140~142	Resolved

Description

`handleSupplyCapExceeding(address)` executes `EBRAKE.pauseSupply(market)` and then `EBRAKE.decreaseCF(market, 0)` in the same transaction after the supply-cap check passes. Because the second call is not isolated with `try/catch`, any revert during collateral-factor reduction reverts the entire transaction and undoes the earlier supply pause.

This creates a real single-point failure in the emergency halt path for markets where the collateral-factor update can revert, including isolated-pool markets.

The failure mode is supported for isolated-pool markets, where the comptroller-side collateral-factor path can revert under conditions such as `MarketNotListed`, `InvalidCollateralFactor`, `InvalidLiquidationThreshold`, or `PriceError`. If such a revert occurs inside `EBRAKE.decreaseCF(market, 0)`, the EVM reverts the full transaction, canceling the already-executed supply pause.

Scenario

1. A market exceeds its `supplyCap`, and an authorized caller invokes `handleSupplyCapExceeding(market)`.
2. `EBRAKE.pauseSupply(market)` succeeds.
3. `EBRAKE.decreaseCF(market, 0)` encounters a downstream revert condition, such as `PriceError` or another isolated-pool collateral-factor validation failure.
4. The entire transaction reverts, rolling back the earlier pause.
5. Supply remains unpaused until a separate successful intervention occurs.

Recommendation

Decouple the safety-critical supply pause from the collateral-factor reduction so a failure in the second step cannot unwind the first. A practical approach is to keep `EBRAKE.pauseSupply(market)` as the primary containment action and wrap `EBRAKE.decreaseCF(market, 0)` in `try/catch`, handling failure explicitly with an event or alternative follow-up path. If atomic execution is intentional, document that `handleSupplyCapExceeding()` is best-effort and that operators may need to perform pause and collateral-factor actions separately when downstream CF updates are not reliable.

Alleviation

[Venus Labs, 04/27/2026]:

The cited revert paths cannot fire for this specific call. `handleSupplyCapExceeding` invokes `decreaseCF(market, 0)`, and `EBrake` forwards `newCF=0` along with the market's current LT to the comptroller's `setCollateralFactor`. Under those arguments:

1. `InvalidCollateralFactor` unreachable (0 less than equal to `MAX_CF`).
2. `InvalidLiquidationThreshold` unreachable (reuses the existing valid LT; LT greater than equal to 0 always).
3. `PriceError` unreachable, explicitly gated on `newCF != 0` in both IL and Diamond comptrollers. Oracle is never queried when `newCF=0`.
4. `MarketNotListed` only possible if a market is delisted after `setMarketConfig`, which is not a pattern Venus uses.

On a successful pause + failed CF update, we would emit the success event and leave CF unchanged on the comptroller, producing an inconsistency between the event log and on-chain state.

APPENDIX | VENUS LABS - EXECUTOR CONTRACT

Finding Categories

Categories	Description
Volatile Code	Volatile Code findings refer to segments of code that behave unexpectedly on certain edge cases and may result in vulnerabilities.

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